

Experiment No. 1

Experiment Name

Implementation of Elementary Discrete-Time Signals Using MATLAB/Python

Definition

A **discrete-time signal** is a signal that is defined only at discrete values of the independent variable, usually represented by n .

Objectives

- To understand elementary discrete-time signals.
- To generate the unit sample signal.
- To generate the unit step signal.
- To generate the unit ramp signal.
- To plot these signals using MATLAB/Python.

Theory

The basic elementary discrete-time signals are:

1. Unit Sample Signal

The unit sample or discrete impulse signal is defined as

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

It has a value of 1 only at $n = 0$.

2. Unit Step Signal

The unit step signal is defined as

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

It changes from 0 to 1 at $n = 0$.

3. Unit Ramp Signal

The unit ramp signal is defined as

$$r(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

It increases linearly with n for $n \geq 0$.

These signals are fundamental building blocks for discrete-time signal analysis.

Experiment No. 2

Experiment Name

Generation and Plotting of Discrete-Time Impulse Sequences

Definition

A **unit impulse signal** is a discrete-time signal that has amplitude 1 at $n = 0$ and zero at all other values of n .

Objectives

- To understand shifted impulse signals.
- To generate the given discrete-time sequence.
- To determine the position and amplitude of each impulse.
- To plot the sequence for $-7 \leq n \leq 7$.

Theory

The given sequence is

$$x(n) = 2\delta(n + 2) - \delta(n - 4)$$

A shifted impulse follows

$$\delta(n - k)$$

where the impulse occurs at $n = k$.

Therefore,

$$\delta(n + 2) = \delta[n - (-2)]$$

has its impulse at

$$n = -2.$$

Similarly,

$$\delta(n - 4)$$

has its impulse at

$$n = 4.$$

Thus,

$$x(-2) = 2$$

and

$$x(4) = -1.$$

The signal is zero at all other values of n .

Experiment No. 3

Experiment Name

Implementation of Basic Discrete-Time Signal Operations Using User-Defined Functions

Definition

Signal operation means performing mathematical transformations on a signal, such as addition, folding, and shifting, to obtain a new signal.

Objectives

- To understand basic discrete-time signal operations.
- To perform addition of two signals.
- To perform folding of a signal.
- To perform shifting of a signal.
- To implement these operations using user-defined functions.
- To plot the original and transformed signals.

✦ Upgrade



Theory

1. Addition

Addition of two discrete-time signals is defined as

$$y(n) = x_1(n) + x_2(n).$$

The corresponding samples of the two signals are added.

2. Folding

Folding is the process of reversing a signal about the vertical axis $n = 0$.

$$y(n) = x(-n)$$

For example, if

$$x(n) = \{1, 2, 3\}$$

then the folded sequence appears in reverse order.

3. Shifting

Time shifting means moving a signal along the time axis.

For a right shift of k ,

$$y(n) = x(n - k).$$

For a left shift of k ,

$$y(n) = x(n + k).$$

Experiment No. 4

Experiment Name

Implementation of Convolution Operation of Discrete-Time Sequences

Definition

Convolution is a mathematical operation used to determine the output of an LTI discrete-time system from its input signal and impulse response.

Objectives

- To understand discrete-time convolution.
- To calculate the convolution of two sequences.
- To implement convolution using MATLAB/Python.
- To verify the convolution result.

- To plot the input, impulse response, and output signals.

Theory

For two discrete-time sequences $x(n)$ and $h(n)$, convolution is defined as

$$y(n) = x(n) * h(n)$$

or

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k).$$

Here,

- $x(n)$ = input sequence
- $h(n)$ = impulse response
- $y(n)$ = output sequence

The basic process of convolution consists of:

1. Folding one sequence.
2. Shifting the folded sequence.
3. Multiplying corresponding samples.
4. Adding all products.

For an LTI system,

$$y(n) = x(n) * h(n)$$

Experiment No. 5

Experiment Name

Generation of Sinusoidal Waves with Different Frequencies Using MATLAB/Python

Definition

A **sinusoidal signal** is a periodic signal whose amplitude varies according to a sine or cosine function.

Objectives

- To understand sinusoidal signals.
- To generate sinusoidal signals with different frequencies.

- To observe the effect of frequency on the waveform.
- To calculate the period of sinusoidal signals.
- To plot the signals using MATLAB/Python.

Theory

A sinusoidal signal is represented by

$$x(t) = A \sin(2\pi ft + \phi)$$

where:

- A = amplitude
- f = frequency
- t = time
- ϕ = phase

Angular frequency is

$$\omega = 2\pi f.$$

The period is

$$T = \frac{1}{f}.$$

If frequency increases, the number of cycles per unit time increases and the period decreases.

Experiment No. 6

Experiment Name

Implementation of Discrete Fourier Transform (DFT) and Inverse Discrete Fourier Transform (IDFT)

Definition

The **Discrete Fourier Transform (DFT)** is a mathematical technique used to convert a finite discrete-time signal from the time domain into the frequency domain.

Objectives

- To understand DFT.

- To calculate the frequency components of a discrete signal.
- To implement DFT using MATLAB/Python.
- To understand IDFT.
- To reconstruct the original signal using IDFT.
- To verify the DFT-IDFT relationship.

Theory

For an N -point sequence $x(n)$, the DFT is

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi kn}{N}}$$

where

$$k = 0, 1, \dots, N - 1.$$

The IDFT is

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi kn}{N}}.$$

DFT provides the frequency-domain representation of a finite discrete signal, while IDFT converts it back to the time domain.

Experiment No. 7

Experiment Name

Determination of the Spectrum of a Multi-Frequency Signal

Definition

The **spectrum of a signal** represents the distribution of its frequency components and their corresponding amplitudes and phases.

Objectives

- To generate a multi-frequency signal.
- To identify the frequency components.
- To determine its spectrum.
- To plot the frequency spectrum.

- To analyze the relationship between time and frequency domains.

Theory

The given signal is

$$x(k) = 0.25 + 2 \sin(2\pi 5k) + \sin(2\pi 12.5k) + 1.5 \sin(2\pi 20k) + 0.5 \sin(2\pi 35k).$$

A multi-frequency signal consists of multiple sinusoidal components with different frequencies.

The spectrum shows the frequencies present in the signal.

The constant component

$$0.25$$

represents the **DC component**.

The sinusoidal components occur at frequencies:

$$5, \quad 12.5, \quad 20, \quad 35$$

with their respective amplitudes.

Experiment No. 8

Experiment Name

Removal of Noise from an Audio Signal Using DFT

Definition

Noise is an unwanted random or deterministic disturbance that affects the quality of a useful signal.

Objectives

- To generate a pure 440 Hz audio signal.
- To add random noise to the signal.
- To transform the noisy signal into the frequency domain.
- To remove unwanted high-frequency components.
- To reconstruct the cleaned signal using IDFT.
- To compare the original, noisy, and cleaned signals.

Theory

A pure audio signal can be represented as

$$x(t) = A \sin(2\pi ft).$$

For this experiment,

$$f = 440 \text{ Hz}.$$

After adding noise,

$$x_{\text{noisy}}(t) = x(t) + n(t)$$

where $n(t)$ represents noise.

The DFT transforms the noisy signal into the frequency domain.

Unwanted frequency components can then be attenuated or removed using a frequency-domain filter.

Finally, IDFT is applied to obtain the cleaned signal:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi kn/N}.$$

Experiment No. 9

Experiment Name

Extraction of Systolic Peaks, Diastolic Points, and Heart Rate from PPG Signal

Definition

Photoplethysmography (PPG) is a non-invasive optical technique that measures changes in blood volume in tissue.

Objectives

- To understand the PPG signal.
- To process a PPG waveform.
- To detect systolic peaks.
- To identify diastolic points.

- To calculate heart rate.
- To plot the extracted PPG features.

Theory

A PPG signal represents changes in blood volume associated with cardiac activity.

Important features include:

Systolic Peak:

The main peak of the PPG waveform corresponding to the rapid increase in blood volume during the cardiac cycle.

Diastolic Point:

A secondary feature occurring after the systolic peak and associated with the later phase of the pulse waveform.

Heart Rate:

The number of cardiac cycles per minute.

If T is the time interval between two consecutive systolic peaks,

$$HR = \frac{60}{T} \text{ BPM.}$$

Thus, detecting consecutive systolic peaks allows the heart rate to be estimated.

Experiment No. 10

Experiment Name

Determination of Power Spectral Density of a Square Wave

Definition

Power Spectral Density (PSD) describes how the power of a signal is distributed over different frequencies.

Objectives

- To generate a square wave.
- To understand Power Spectral Density.
- To calculate the PSD of the square wave.

- To plot the frequency spectrum.
- To observe the harmonic components.

Theory

An ideal square wave can be represented by the Fourier series

$$x(t) = \frac{4}{\pi} \left[\sin(\omega_0 t) + \frac{1}{3} \sin(3\omega_0 t) + \frac{1}{5} \sin(5\omega_0 t) + \dots \right].$$

Therefore, a square wave contains:

$$f_0, 3f_0, 5f_0, 7f_0, \dots$$

That means a square wave mainly contains the **fundamental frequency and odd harmonics**.

PSD shows how the signal power is distributed among these frequency components.

Experiment No. 11

Experiment Name

Fourier Series Approximation of a Square Wave

Definition

A **Fourier series** is a mathematical representation of a periodic signal as a sum of sinusoidal signals with different frequencies, amplitudes, and phases.

Objectives

- To understand Fourier series.
- To generate a square wave using Fourier series.
- To approximate a square wave using finite harmonics.
- To observe the effect of increasing harmonics.
- To compare the approximation with the ideal square wave.

Theory

The Fourier series of a square wave is

$$x(t) = \frac{4}{\pi} \left[\sin(\omega_0 t) + \frac{1}{3} \sin(3\omega_0 t) + \frac{1}{5} \sin(5\omega_0 t) + \frac{1}{7} \sin(7\omega_0 t) + \dots \right].$$

In general,

$$x(t) = \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n} \sin(n\omega_0 t).$$

If only a limited number of terms are used, the result is an approximation.

As the number of harmonics increases, the approximation becomes closer to the ideal square wave.

Experiment No. 12

Experiment Name

Determination of Amplitude Spectrum of a Multi-Frequency Signal

Definition

An **amplitude spectrum** represents the magnitude of the frequency components present in a signal as a function of frequency.

Objectives

- To generate the given multi-frequency signal.
- To determine its amplitude spectrum.
- To identify the frequency components.
- To plot the amplitude spectrum.
- To analyze the signal for $0 \leq f \leq 900$ Hz.

Theory

The given signal is

$$x(t) = \cos(2\pi 100t) + \cos(2\pi 500t) + \cos(2\pi 700t).$$

The signal contains three frequency components:

$$100 \text{ Hz}, \quad 500 \text{ Hz}, \quad 700 \text{ Hz}.$$

For a cosine signal,

$$x(t) = A \cos(2\pi f_0 t),$$

the spectrum contains frequency components at

$$+f_0 \quad \text{and} \quad -f_0.$$

Therefore, the amplitude spectrum shows peaks at the corresponding frequencies.

The amplitude spectrum helps identify the dominant frequency components of a signal.

Experiment No. 13

Experiment Name

Determination of the Discrete-Time Fourier Transform (DTFT) of a Finite-Duration Sequence

Definition

The **Discrete-Time Fourier Transform (DTFT)** is a mathematical transformation that converts a discrete-time signal into its frequency-domain representation.

Objectives

- To understand the concept of DTFT.
- To calculate the DTFT of a finite-duration sequence.
- To determine the magnitude spectrum.
- To determine the phase spectrum.
- To plot the DTFT for $0 \leq \omega \leq \pi$.
- To evaluate the DTFT at 501 equally spaced frequency points.

Theory

The DTFT of a discrete-time signal $x(n)$ is

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}.$$

For the given sequence,

$$x(n) = \{1, 2, 3, 4, 5\}$$

assuming it starts at $n = 0$,

$$x(0) = 1, \quad x(1) = 2, \quad x(2) = 3, \quad x(3) = 4, \quad x(4) = 5.$$

Therefore,

$$X(e^{j\omega}) = 1 + 2e^{-j\omega} + 3e^{-j2\omega} + 4e^{-j3\omega} + 5e^{-j4\omega}.$$

The magnitude spectrum is

$$|X(e^{j\omega})|$$

and the phase spectrum is

$$\angle X(e^{j\omega}).$$

The DTFT is periodic with period

$$2\pi.$$

For this experiment, the DTFT is evaluated at **501 equally spaced frequency points** from 0 to π .

Important Definitions for Viva

Term	Definition
Signal	A function that carries information about a physical phenomenon.
Discrete-Time Signal	A signal defined only at discrete values of time/index n .
Unit Impulse	A signal having value 1 at $n = 0$ and 0 elsewhere.
Unit Step	A signal having value 1 for $n \geq 0$ and 0 for $n < 0$.
Unit Ramp	A signal whose amplitude increases linearly with n for $n \geq 0$.
Folding	Reversal of a signal about $n = 0$, $x(n) \rightarrow x(-n)$.
Shifting	Moving a signal along the time/index axis.
Convolution	An operation used to determine the output of an LTI system from input and impulse response.
Sinusoidal Signal	A periodic signal represented by sine or cosine function.

Term	Definition
DFT	Converts a finite discrete signal from time domain to frequency domain.
IDFT	Converts the DFT representation back to the time domain.
DTFT	Gives the frequency-domain representation of a discrete-time signal over continuous frequency.
Spectrum	Representation of signal components with respect to frequency.
Amplitude Spectrum	Shows the magnitude of signal components versus frequency.
PSD	Shows how signal power is distributed over frequency.
Noise	An unwanted component that distorts a useful signal.
Fourier Series	Represents a periodic signal as a sum of sinusoidal components.
PPG	Optical technique for measuring changes in blood volume in tissue.
Systolic Peak	Main peak of a PPG pulse waveform.
Heart Rate	Number of cardiac cycles per minute, expressed in BPM.



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